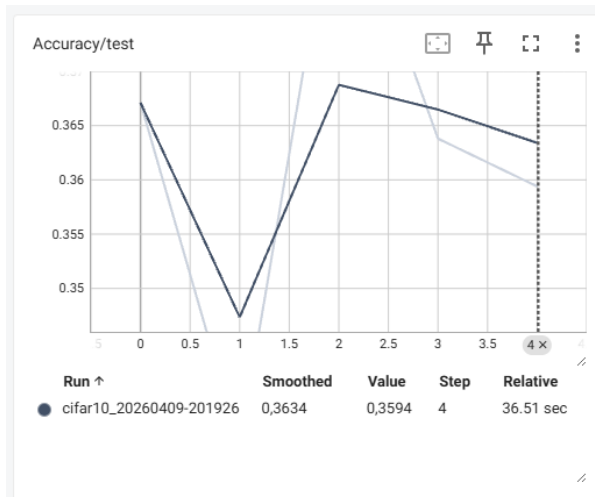


I wanted to challenge myself with a new dataset, so I chose the CIFAR(10) dataset. I've also added the script in my Github, so you can see how I apply my logic throughout the script. I started off by implementing a simple baseline model to ensure proper training, tuning and logging. Baseline model has poor accuracy (as would be expected), 0.3594* after 5 epochs. (*Note added later: I noticed in my later runs that my outcomes were not deterministic, so I implemented a fixed seed to ensure proper testing from there on out. This is why the value of the screenshot is different from my table below)



My hypothesis is: *Increasing the complexity of my model will improve classification accuracy on CIFAR-10, up to a point where performance flips due to overfitting.*

I've done various improvements, I've added a table below that shows the results per improvement.

Model	Improvement	Accuracy	Comments
Baseline	None, baseline model.	0.3829	Epochs = 5
Improved_v1	Conv2d model	0.3966	Still linear
Improved_v2	Conv2d non linear	0.5587	Added ReLU
Improved_v3	Raised hidden units to 32 (doubled)	0.5969	
Improved_v4	Added a layer	0.6297	Learning curve still steep, so will increase epochs
Improved_v5	Increased epochs to 10	0.6049	Accuracy actually decreased, I will add pooling now
Improved_v6	Added pooling	0.6675	Epochs to 5 again for a bit faster training
Improved_v7	Epochs to 10 again to see if it helps...	0.6823	
Improved_v8	Added dropout	0.6525	

Improved_v9	Added another layer	0.6998	
Improved_v10	Added BatchNorm	0.6289	Steady increase until last epoch, then a massive dip from 0.7085 to 0.6289. Overall over the epochs, accuracy improved so improvement seem to have been useful. I will continue training for now with different improvements but will keep this in mind.
Improved_v11	Increased amount of channels	0.7743	Training seems stable now.
Improved_v12	Adding classifier head (before final classification)	0.7785	One run had 0.7924
Improved_v13	Added random horizontal flip to training	0.7745	
Improved_v14	Added random crop	0.7745	Other epochs had different accuracy, so seems like a coincidence that accuracy is the same
Improved_v15	Increased epochs to 20	0.7731	But training loss improved nice and steady. Model seems to be quite stable at this point.

The readme shows that there was a 18% test error. My model reached 77.31% accuracy, which I think comes close already to the mentioned test error with optimal hyperparameter tuning. Below I've added the curves of training as visualized via TensorBoard.

